

# README: Silicon Radiation Damage Project

March 20, 2026

This project studies the chain

radiation  $\rightarrow$  defects in silicon  $\rightarrow$  material degradation  $\rightarrow$  device degradation.

The repository is divided into:

- Geant4 radiation transport module
- MATLAB defect and device degradation module
- notes and derivations for presentation preparation

The intended workflow is:

1. run Geant4 to produce radiation-interaction outputs,
2. use MATLAB to convert those outputs into defect-density and device-metric trends,
3. use the notes and derivations to support slides and discussion.

## Main physics idea

Energetic particles can transfer enough energy to displace silicon atoms from lattice sites. The resulting vacancies, interstitials, and defect complexes create trap and recombination states that reduce carrier lifetime and increase leakage current.

## Cryogenic extension

An optional extension studies how the kinetics of radiation-induced defects change as temperature is lowered from room temperature toward the cryogenic regime. The extension models:

- defect migration freeze-out,
- trap emission freeze-out,
- annealing freeze-out.

The corresponding note files are `docs/physics_notes/cryogenic_defect_evolution.tex` and `docs/derivations`. The MATLAB entry point for this section is `matlab/main_cryogenic_extension.m`.